# INS 401: Project Management

## Managing Project Quality

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8 December 2025
Main Campus, Lagos

# Ground Rules

- Participate
- Respect each other
- Silence Phones
- Be on time

# Outline

- Definition and dimensions of project quality
- Tools and techniques for managing quality

# Learning Outcomes

At the end of this class, students should be able to:

- describe project quality and its dimensions
- discuss the tools and techniques for managing project quality

# Project Quality – What is it?

- Quality is the degree to which a set of inherent characteristics fulfills requirements – *PMI PMBOK Guide*

- Project quality may be defined as the degree to which a project's deliverables and outcomes meet stakeholder requirements and expectations.

# Project Quality Management

Project Quality Management includes the processes for incorporating the organization's quality policy regarding planning, managing and controlling project and product quality requirements in order to meet stakeholder objectives.

Project Quality Management processes include the following:

- Plan Quality Management
- Manage Quality
- Control Quality

# PMI Process Groups and Knowledge Areas Mapping

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## PROCESS GROUP AND KNOWLEDGE AREA MAPPING

| Knowledge Areas | Project Management Process Groups | | | | |
|---|---|---|---|---|---|
| | Initiating | Planning | Executing | Monitoring and Controlling | Closing |
| 4. Project Integration Management | 4.1 Develop Project Charter | 4.2 Develop Project Management Plan | 4.3 Direct and Manage Project Work<br>4.4 Manage Project Knowledge | 4.5 Monitor and Control Project Work<br>4.6 Perform Integrated Change Control | 4.7 Close Project or Phase |
| 5. Project Scope Management | | 5.1 Plan Scope Management<br>5.2 Collect Requirements<br>5.3 Define Scope<br>5.4 Create WBS | | 5.5 Validate Scope<br>5.6 Control Scope | |
| 6. Project Schedule Management | | 6.1 Plan Schedule Management<br>6.2 Define Activities<br>6.3 Sequence Activities<br>6.4 Estimate Activity Durations<br>6.5 Develop Schedule | | 6.6 Control Schedule | |
| 7. Project Cost Management | | 7.1 Plan Cost Management<br>7.2 Estimate Costs<br>7.3 Determine Budget | | 7.4 Control Costs | |
| 8. Project Quality Management | | 8.1 Plan Quality Management | 8.2 Manage Quality | 8.3 Control Quality | |
| 9. Project Resource Management | | 9.1 Plan Resource Management<br>9.2 Estimate Activity Resources | 9.3 Acquire Resources<br>9.4 Develop Team<br>9.5 Manage Team | 9.6 Control Resources | |
| 10. Project Communications Management | | 10.1 Plan Communications Management | 10.2 Manage Communications | 10.3 Monitor Communications | |
| 11. Project Risk Management | | 11.1 Plan Risk Management<br>11.2 Identify Risks<br>11.3 Perform Qualitative Risk Analysis<br>11.4 Perform Quantitative Risk Analysis<br>11.5 Plan Risk Responses | 11.6 Implement Risk Responses | 11.7 Monitor Risks | |
| 12. Project Procurement Management | | 12.1 Plan Procurement Management | 12.2 Conduct Procurements | 12.3 Control Procurements | |
| 13. Project Stakeholder Management | 13.1 Identify Stakeholders | 13.2 Plan Stakeholder Engagement | 13.3 Manage Stakeholder Engagement | 13.4 Monitor Stakeholder Engagement | |

Source: *PMBOK® Guide*, 6th Edition

# Plan Quality Management

Plan Quality Management is the process of identifying quality requirements and/or standards for the project and its deliverables, and documenting how the project will demonstrate compliance with quality requirements and/or standards. It provides guidance and direction on how quality will be managed and verified throughout the project.

**Key Output**

- **Quality management plan** – a component of the project management plan that describes how applicable policies, procedures and guidelines will be implemented to achieve the quality objectives.

- **Quality metrics** – a quality metric specifically describes a project or product attribute and how Control Quality process will verify compliance to it. Examples include errors found per line code, failure rate, etc.

# Plan Quality Management

| Inputs | Tools & Techniques | Outputs |
|---|---|---|
| 1. Project charter<br>2. Project management plan<br>-Requirements management plan<br>-Risk management plan<br>-Stakeholder engagement plan<br>-Scope baseline<br>3. Project documents<br>-Assumption log<br>-Requirements documentation<br>-Requirements traceability matrix<br>-Stakeholder register<br>-Risk register<br>4. Enterprise Environmental factors<br>5. Organizational process assets | 1. Expert judgement<br>2. Data gathering<br>-Brainstorming<br>-Interviews<br>-Benchmarking<br>3. Data analysis<br>-Cost-benefit analysis<br>-Cost of quality<br>4. Decision making<br>-Multicriteria decision analysis<br>5. Data representation<br>-Flowcharts<br>-Matrix diagrams<br>-Mind mapping<br>6. Test and inspection planning<br>7. Meetings | 1. Quality management plan<br>2. Quality metrics<br>3. Project management plan updates<br>-Risk management plan<br>-Scope baseline<br>4. Project documents updates<br>-Lessons learned register<br>-Requirements traceability matrix<br>-Risk register<br>-Stakeholder register |

# Manage Quality

Manage Quality is the process of translating the quality management plan into executable quality activities that incorporate the organization's quality policies into the project. Manage quality uses the data and results from the control policy process to reflect the overall quality status of the project to the stakeholders.

**Core outputs**

- **Quality reports** – a quality report is a project document that includes quality management issues, recommendations for corrective actions, and a summary of findings from quality control activities and may include recommendations for process, project and product improvements.
- **Test and evaluation documents** – project documents that describe the activities used to determine if the product meets the quality objectives stated in the quality management plan.
- **Change requests**

# Manage Quality

## Inputs

1. Project management plan
   - -Quality management plan
2. Project documents
   - -Lessons learned register
   - -Quality control measurements
   - -Quality metrics
   - -Risk report
3. Organizational process assets

## Tools & Techniques

1. Data gathering
   - -Checklists
2. Data analysis
   - -Alternative analysis
   - -Document analysis
   - -Process analysis
   - -Root cause analysis
3. Decision making
   - -Multicriteria decision analysis
4. Data representation
   - -Affinity diagrams
   - -Matrix diagrams
   - -Cause-and-effect diagrams
   - -Flowcharts
5. Audits
6. Problem solving
7. Quality improvement methods

## Outputs

1. Quality reports
2. Test and evaluation documents
3. Change requests
4. Project management plan updates
   - -Quality management plan
   - -Scope baseline
   - -Schedule baseline
   - -Cost baseline
5. Project documents updates
   - -Issue log
   - -Lessons learned register
   - -Risk register

# Control Quality

Control Quality is the process of monitoring and recording results of executing the quality management activities in order to assess performance and ensure the project outputs are complete, correct and meet customer expectations.

**Core outputs**

- **Quality control measurements** – these are the documented results of Control Quality activities.
- **Verified deliverables** – completed project deliverables that have been checked and confirmed for correctness through the Control Quality process. objectives stated in the quality management plan. These deliverables become an input to the Validate Scope process for formalized acceptance.
- **Work performance information** –includes information on project requirements fulfilment, causes for rejections, reworks required, lists of verified deliverables and the need for process adjustments.
- **Change requests**

# Control Quality

## Inputs

1. Project management plan
   -Quality management plan
2. Project documents
   -Lessons learned register
   -Quality metrics
   -Test and evaluation documents
3. Approved change requests
4. Deliverables
5. Work performance data
6. Enterprise environmental factors
7. Organizational process assets

## Tools & Techniques

1. Data gathering
   -Checklists
   -Check sheets
   -Statistical sampling
   -Questionnaires and surveys
2. Data analysis
   -Performance reviews
   -Root cause analysis
3. Inspection
4. Testing/product evaluations
5. Data representation
   -Cause-and-effect diagrams
   -Control charts
   -Histogram
   -Scatter diagrams
6. Meetings

## Outputs

1. Quality control measurements
2. Verified deliverables
3. Change requests
4. Project management plan updates
   -Quality management plan
5. Project document updates
   -Issue log
   -Lessons learned register
   -Risk register
   -Test and evaluation documents

# Tools and Techniques for Managing Project Quality

There are several tools and techniques used for managing project quality. They include:

**Cause and Effect Diagram** – a decomposition technique that helps trace an undesirable effect back to its root cause. Also called fishbone diagram.

**Benchmarking –** the comparison of actual or planned products, processes and practices to those of comparable organizations to identify best practices, generate ideas for improvement, and provide a basis for measuring performance.

**Checklists –** a checklist is a list of items, actions or points to be considered. It can be useful in managing the control quality activities in a structured manner.

**Pareto Charts -** Pareto charts are used in quality control to assess the most frequently occurring defects by category.

**Six Sigma –** see details in subsequent slides

**Cost of Quality** - see details in subsequent slides

# Cost of Quality [1]

The cost of quality associated with a project can be categorized into the following costs:

- **Prevention costs** – costs related to the prevention of poor quality in the products, deliverables or services of the specific project.

- **Appraisal costs** – costs related to evaluating, measuring, auditing and testing the products, deliverables or services of the specific project.

- **Failure costs** – (internal/external) – costs related to non-conformance of the products, deliverables or services to the needs or expectations of the stakeholders.

# Cost of Quality [2]

## Cost of Conformance

**Prevention Costs**
(Build a quality product)

- Training
- Document processes
- Equipment
- Time to do it right

**Appraisal Costs**
(Assess the quality)

- Testing
- Destructive testing loss
- Inspections

## Cost of Nonconformance

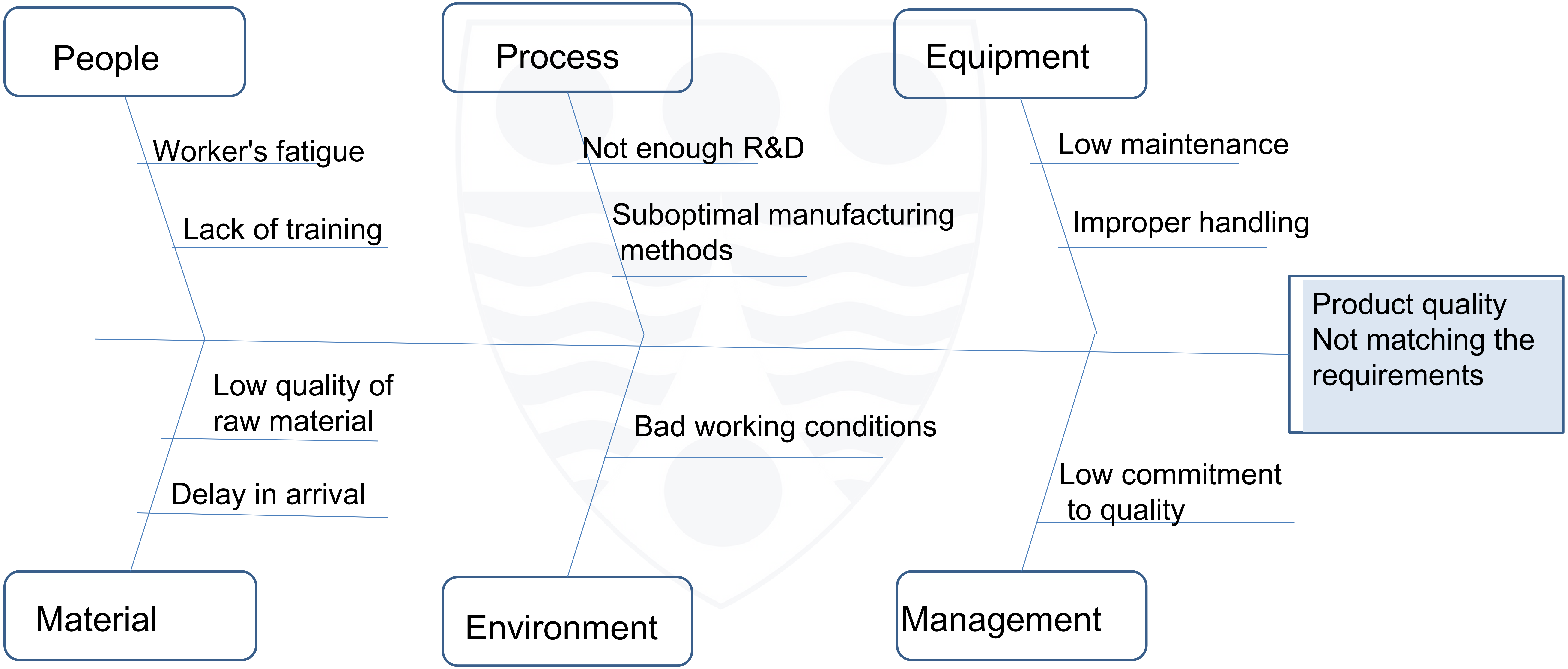
**Internal Failure Costs**
(Failures found by the project)

- Rework
- Scrap

**External Failure Costs**
(Failures found by the customer)

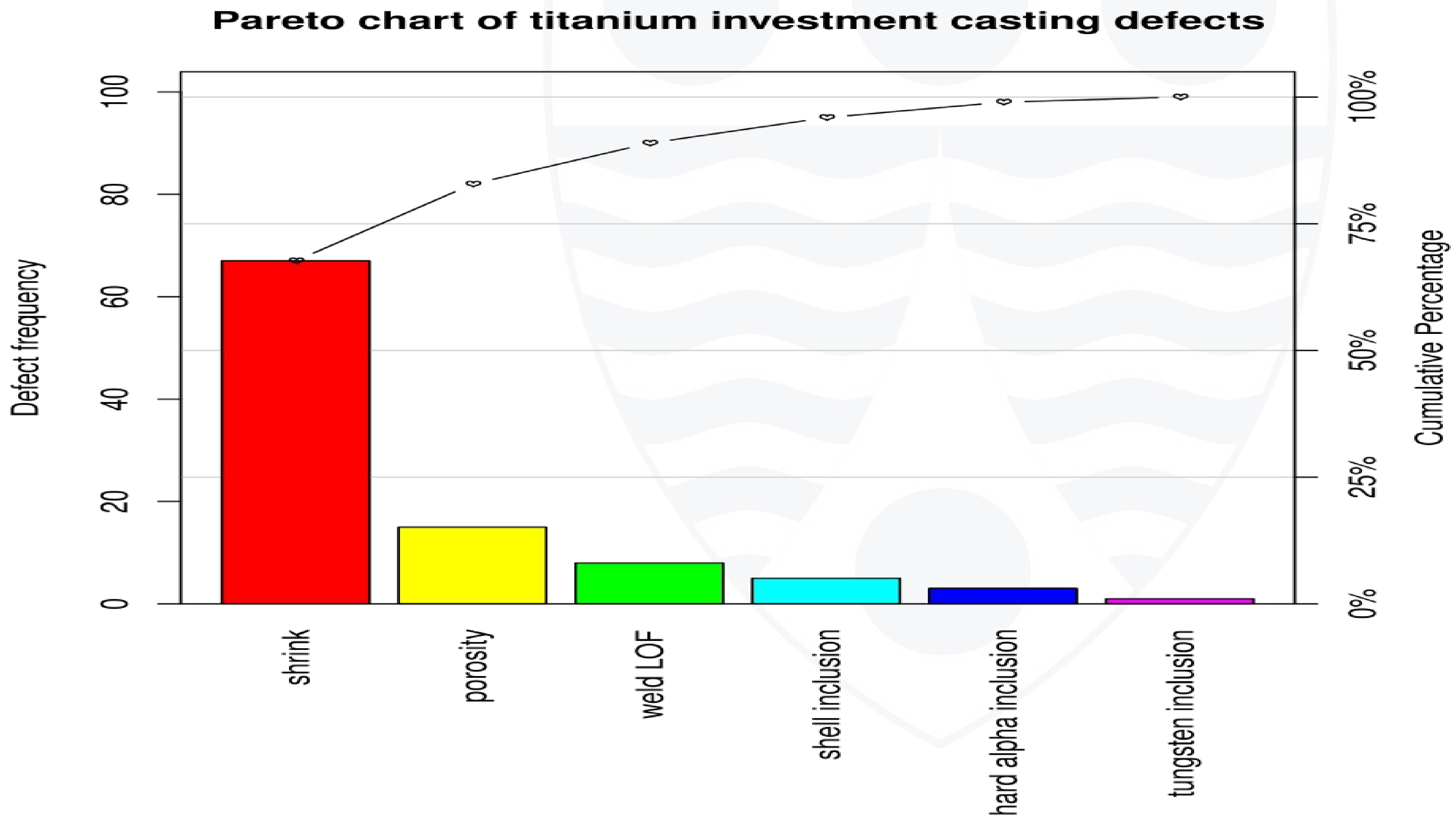
- Liabilities
- Warranty work
- Lost business

# Cause-and-Effect Diagram



# Pareto Chart

Pareto charts are used in quality control to assess the most frequently occurring defects by category.




Source: Wikipedia

# Six Sigma Quality Management Technique

Six Sigma is a quality control methodology that businesses use to eliminate waste and defects in products and services and improve processes. It uses a step-by-step approach called DMAIC – Define, Measure, Analyze, Improve and Control.

Six Sigma quality is achieved when long-term defect levels are below 3.4 defects per Million opportunities (DPMO).

# Six Sigma DMAIC Methodology

Six Sigma DMAIC methodology has five phases.

**Define** – the team chooses a process to focus on and defines the problem it wishes to solve.

**Measure** – the team measures key aspects of the current process, collects relevant data (which serves as a benchmark) and pinpoints a list of inputs that may be hindering performance.

**Analyze** – The team analyzes the process by isolating each input, or potential reason for any failures, and testing it as the possible root cause of the problem. Data is analysed to establish cause and affect and to seek out the root cause of the defect.

**Improve** - The team works to implement changes that will improve system performance.

**Control** - The team adds controls to the process to ensure it does not regress and become ineffective once again.

# Conclusion

- Project quality management applies to all projects, regardless of the nature of the deliverables.

- Quality measures and techniques are specific to the type of deliverables being produced by the project.

# Closing Remarks